# Week 10 Notes

STAT 251 SECTION 01

# From estimating to testing

For the last two weeks we have been **estimating** parameters: a confidence interval answers the question *"what is $p$ (or $\mu$), give or take?"* A lot of questions are not asked that way. They come as a **claim**, and we want to know whether the data back it up:

- A new drug claims to lower blood pressure. Does it, or did the patients' readings just happen to drop?
- The polling averages say a Senate candidate has 52% support. A new poll finds less. Has support really dropped, or is the new poll just a different sample?
- Your roommate says they can tell Coke from Pepsi. Can they, or are they guessing?

Each claim is about a **parameter**: the drug's average effect $\mu$, the candidate's support $p$, your roommate's chance $p$ of naming a cup correctly. We never see the parameter. We see an **estimate** from a sample, and estimates bounce around from sample to sample (Week 8: two polls of the same state give two different answers). So the real question is always:

> **Is the difference between what we saw and what the claim says too big to be ordinary sampling variation?**

A **significance test** (also called a **hypothesis test**) is the tool that answers it. A result that is decidedly **not** due to ordinary variation, that is, not due to chance or coincidence, is called **statistically significant**.

# The logic of a significance test

## In plain words: a courtroom

A significance test works like a criminal trial.

- The defendant is **presumed innocent**. Nobody has to prove innocence; innocence is the starting assumption. In a test, the starting assumption is the boring explanation, "nothing

interesting is going on", and it is called the **null hypothesis**.

- The prosecution has a **suspicion**: the defendant did it. In a test, the suspicion is called the **alternative hypothesis**. The burden of proof is on the suspicion, not on the boring explanation.
- The jury asks: **if the defendant really were innocent, how likely is evidence like this?** If that is very unlikely (beyond reasonable doubt), they convict. In a test, that probability is the **P-value**, and "beyond reasonable doubt" is a cutoff we pick in advance, the **significance level** $\alpha$.
- A jury that does not convict says **"not guilty"**, never "innocent". Weak evidence does not prove innocence. In a test, we either **reject** the null hypothesis or **do not reject** it. We never "accept" it.

Hold on to that picture. Everything below is the same four ideas with numbers attached.

# Working it through: can you taste the difference?

Your roommate claims they can tell Coke from Pepsi. You set up a blind test: 8 times, you flip a coin to decide which soda to pour, and they taste the cup and name it. They get **all 8 right**. Impressive, or lucky?

## Step 1: name the parameter, and the two stories

The parameter is $p$, your roommate's true, long-run chance of naming a cup correctly. We cannot see $p$; we only see this one test, where they got $\hat{p} = 8/8 = 1$. There are two stories about $p$:

- **Story 1, "just guessing":** they cannot really tell, so each cup is a coin flip, and $p = 0.5$. This is the **null hypothesis**, written $H_0 : p = 0.5$.
- **Story 2, "they can taste it":** they do better than guessing, $p > 0.5$. This is the **alternative hypothesis**, written $H_A : p > 0.5$.

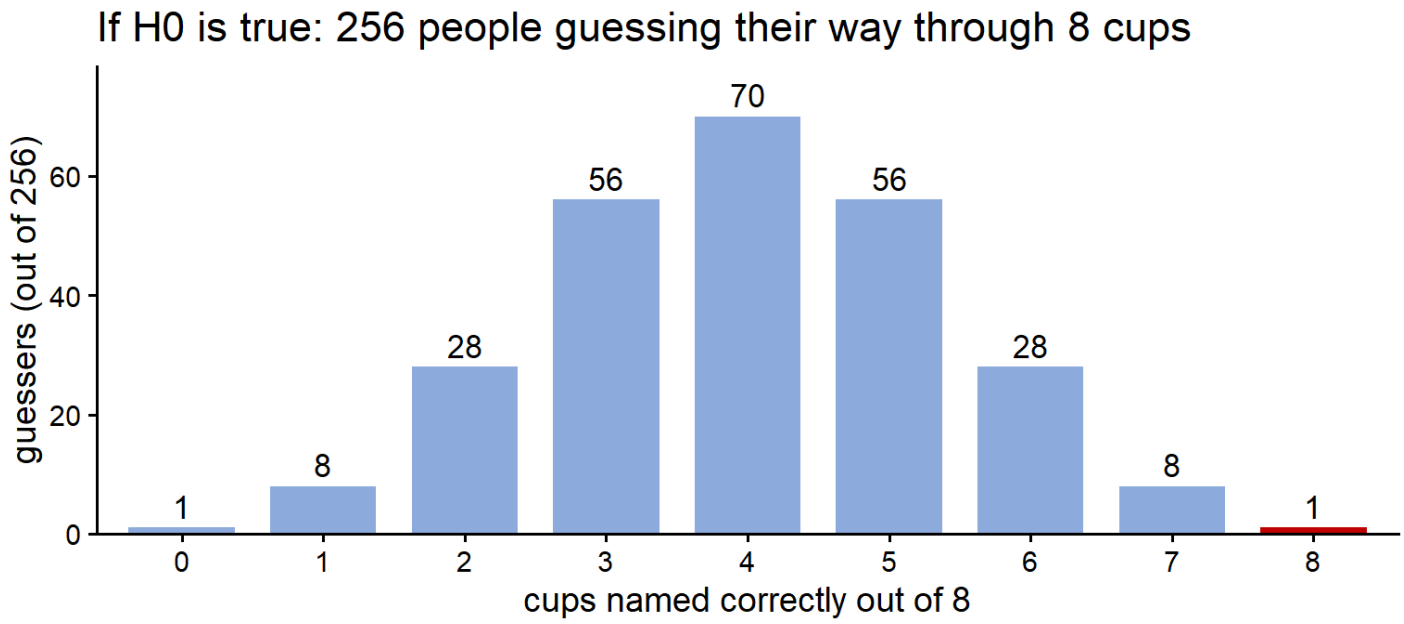
Notice that the hypotheses are about $p$, the parameter, not about $\hat{p}$. We already **know** $\hat{p} = 1$; there is nothing to test about it. The question is what it tells us about $p$.

## Step 2: suppose $H_0$ is true. What results would we expect?

Here is the key move. We cannot prove story 2 directly, so we **assume story 1** and work out what it predicts. If they are only guessing, then the number of cups they get right, $X$, is a count of successes in 8 independent trials that each succeed with probability 0.5. That is a binomial random variable from Week 7:

$$X \sim \text{Binom}(8,\ 0.5) \qquad \text{when } H_0 \text{ is true}$$

A helpful way to see this distribution: imagine **256 people who cannot taste the difference** each take the 8-cup test, guessing every time. Since $2^8 = 256$, the binomial formula $P(X = k) = \binom{8}{k}(0.5)^k(0.5)^{8-k} = \binom{8}{k}/256$ says that, on average, exactly $\binom{8}{k}$ of them score $k$:




This is what "just guessing" looks like. Most guessers land near 4 out of 8, which makes sense: the mean of a binomial is $np = 8(0.5) = 4$. Some get 5 or 6 by luck. Getting **all 8** happens to about **1 guesser in 256**.

## Step 3: how surprising is what we actually saw?

Now compare the data with the prediction. The probability, **if $H_0$ is true**, of a result at least as extreme as the one we saw is called the **P-value**. "At least as extreme" means at least as far in the direction $H_A$ points to, which here is "as many correct, or more":

$$\text{P-value} = P(X \geq 8 \mid H_0 \text{ true}) = \binom{8}{8}(0.5)^8 = \frac{1}{256} \approx 0.0039$$

The bar $\mid$ is read **"given"**: the probability is worked out in the world where $H_0$ is true. That is the same conditional probability idea from Week 6.

A P-value of 0.0039 says: **if your roommate were only guessing, a result this good would happen less than half a percent of the time.**

## Step 4: two explanations, and a decision

There are only two ways to explain a result with a small P-value:

1. **Something unusual happened.** Your roommate was guessing and got very lucky: they are the 1 in 256.
2. **The assumption behind the calculation is wrong.** $H_0$ is false; they really can taste the difference.

Both are possible. The test does not tell us which one is true. It tells us how hard we would have to believe in luck to keep believing $H_0$. The smaller the P-value, the harder that gets, and at some point explanation 2 becomes the more reasonable one.

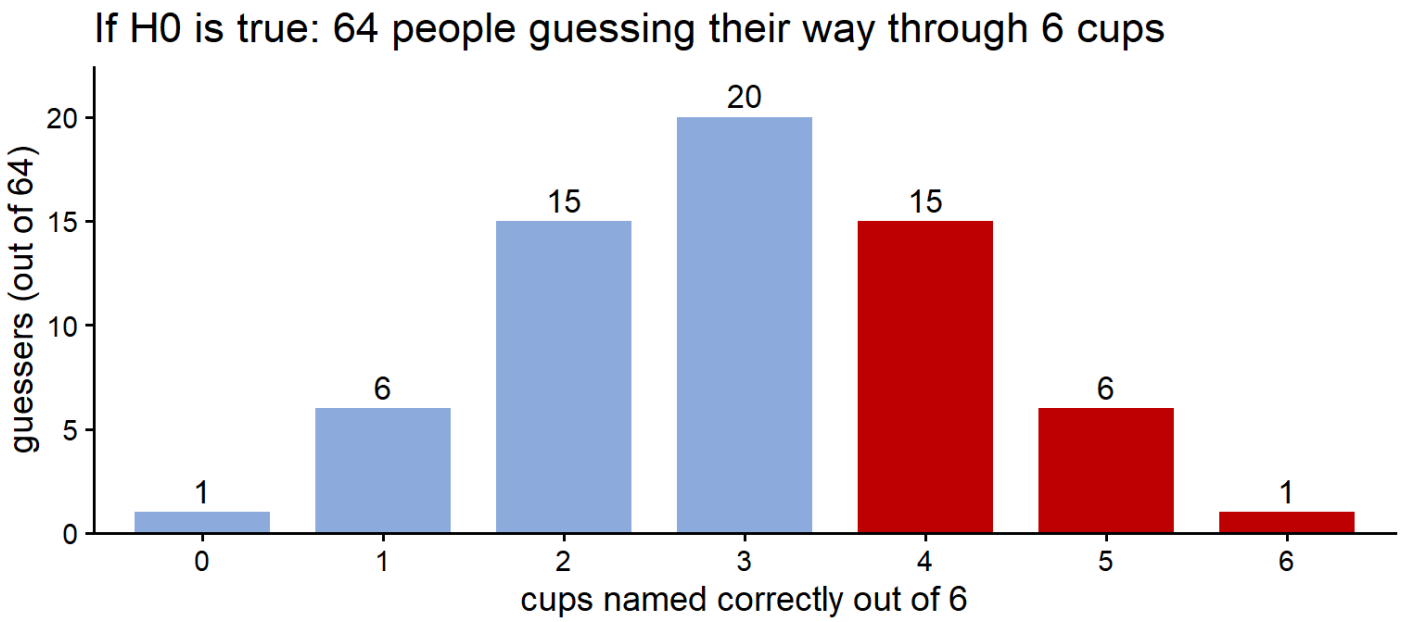
Where is that point? It is a choice, made **before** looking at the data, called the **significance level** $\alpha$. By convention $\alpha = 0.05$, i.e. "if a result this extreme would happen less than 5% of the time by luck alone, we stop believing it was luck". The decision rule is:

- If the P-value $< \alpha$: **reject** $H_0$. The result is statistically significant; the data are evidence for $H_A$.
- If the P-value $\geq \alpha$: **do not reject** $H_0$. The result is not statistically significant; the data are not strong enough evidence for $H_A$.

Here $0.0039 < 0.05$, so we reject $H_0$: the data are evidence that your roommate can taste the difference. The textbook's memory device: **"if the P-value is low, the null must go."**

## A second taster

A friend makes the same claim, but takes a shorter 6-cup test and gets **4 of 6** right. Under $H_0 : p = 0.5$, their score is $X \sim \text{Binom}(6,\ 0.5)$. Picture 64 guessers ($2^6 = 64$):




"At least as extreme" as 4 correct means 4, 5 or 6 correct, the shaded bars:

$$\text{P-value} = P(X \geq 4 \mid H_0 \text{ true}) = \frac{15+6+1}{64} = \frac{22}{64} \approx 0.34$$

About a third of pure guessers do this well, so 4 of 6 is not surprising at all if your friend is guessing. $0.34 \geq 0.05$: **do not reject** $H_0$. This does **not** prove your friend is guessing ("not guilty", not "innocent"). It says this test gives no real evidence that they can taste the difference. A longer test might.

One more thought. If all 256 guessers from the first picture took the 8-cup test, about one of them would score 8 for 8 and be convinced they have a gift. A small P-value means **either** rare luck **or** a false $H_0$, and from one test we cannot be sure which. That is why surprising results get repeated before anyone believes them.

# The notation

Significance tests bring a lot of new symbols. Most of them are old ideas with new names.

| Symbol | Read as | Meaning |
|---|---|---|
| $\theta$ | "theta" | a parameter, any parameter: stands for $p$ or $\mu$ when we talk about tests in general |
| $\theta_0$, e.g. $p_0$, $\mu_0$ | "theta-naught", "p-naught", "mu-naught" | the value of the parameter that $H_0$ claims, a number like 0.5 or 28 g |

| Symbol | Read as | Meaning |
|---|---|---|
| $\hat{\theta}$, e.g. $\hat{p}$, $\bar{x}$ | "theta-hat", "p-hat", "x-bar" | the estimate of $\theta$ from our sample (Weeks 1 and 8) |
| $H_0 : \theta = \theta_0$ | "H-naught: theta equals theta-naught" | the null hypothesis: "no effect", the parameter equals the claimed value |
| $H_A : \theta > \theta_0$ | "H-A: theta is greater than theta-naught" | an upper-tailed alternative: we suspect the parameter is bigger |
| $H_A : \theta < \theta_0$ | "H-A: theta is less than theta-naught" | a lower-tailed alternative: we suspect the parameter is smaller |
| $H_A : \theta \neq \theta_0$ | "H-A: theta is not equal to theta-naught" | a two-sided alternative: we suspect it is different, in either direction |
| $z_{obs}$ | "z observed" | the test statistic: how many standard errors $\hat{\theta}$ is from $\theta_0$ |
| $P(Z \geq z_{obs} \mid H_0 \text{ true})$ | "the probability Z is at least z observed, given H-naught is true" | the P-value of an upper-tailed test |
| $\alpha$ | "alpha" | the significance level, the cutoff for the P-value, usually 0.05 |

The subscript 0 always marks "the value from the null hypothesis". In the taste test, $\theta$ is $p$, $\theta_0 = p_0 = 0.5$, and $\hat{\theta} = \hat{p}$.

# Writing the hypotheses

Hypotheses come first, before any data are looked at. A good order to write them in:

1. **Start with $H_A$**: what do we suspect, or hope to find evidence for? "They can taste the difference" becomes $H_A : p > 0.5$.
2. **Then write $H_0$** as "that effect is not there", with an equals sign: $H_0 : p = 0.5$.

Rules that always hold:

- Both hypotheses are about a **parameter** ($p$ or $\mu$), never an estimate. $H_0 : \hat{p} = 0.5$ makes no sense: we already know $\hat{p}$.
- $H_0$ always contains the **equals sign**, and $H_A$ never does.

- $H_A$ must come from the question, **not from the data**. Looking at the data first and then pointing $H_A$ the way the data went is a form of bias: it makes the evidence look stronger than it is.

The direction of $H_A$ decides which kind of test it is:

| Alternative | Name | Use it when the question says |
|---|---|---|
| $H_A : \theta > \theta_0$ | upper-tailed (one-sided) | "more than", "better", "increased", "at least" |
| $H_A : \theta < \theta_0$ | lower-tailed (one-sided) | "less than", "worse", "decreased", "fewer" |
| $H_A : \theta \neq \theta_0$ | two-sided | "different", "changed", "is it fair?" |

## Example: stating hypotheses

For each story, decide what the parameter is, write $H_A$ first (the suspicion), then $H_0$ (no effect), give $\theta_0$ and say which tail.

a. *The taste test.* Your roommate swears they can tell Coke from Pepsi with their eyes closed. You decide to put it to the test: behind their back, you flip a coin to choose which soda goes in each cup, hand them the cups one at a time, and write down whether they name each one correctly. Someone who is just guessing would be right about half the time. Does your roommate do better than guessing?

*Answer:* $p$ = their chance of naming a cup right. $H_A : p > 0.5$, $H_0 : p = 0.5$, with $p_0 = 0.5$: **upper-tailed**. Only "better than guessing" would support the claim.

b. *The spinning coin.* A friend says that a penny spun on its edge on a table does not land heads half the time, the way a flipped penny does. Neither of you knows which way it should lean: the two faces are stamped differently, so the coin might fall toward heads or toward tails. You plan to spin it many times and count the heads. Is a spun penny fair?

*Answer:* $p$ = the chance a spun penny lands heads. We do not suspect a direction, only "not fair": $H_A : p \neq 0.5$, $H_0 : p = 0.5$, with $p_0 = 0.5$: **two-sided**.

c. *The chip bag.* Every bag of your favorite chips says "Net weight 28 g". Lately the bags seem to have more air and fewer chips, and you suspect the company is quietly under-filling them. You buy a bag from several different stores and weigh the chips in each one. Do the bags hold less than 28 g on average?

*Answer:* $\mu$ = the mean weight of chips in all bags of this brand. $H_A : \mu < 28$, $H_0 : \mu = 28$, with $\mu_0 = 28$ g: **lower-tailed**. Only an under-filled bag would cheat the customer.

d. *The coffee machine.* The campus coffee shop's machine is set to pour 12 oz into a medium cup. If it pours too little, customers complain; if it pours too much, the shop wastes coffee and cups overflow. Each month the manager measures a sample of medium pours to check that the machine is still set correctly. Is the machine's mean pour off from 12 oz?

*Answer:* $\mu$ = the mean amount the machine pours into a medium cup. A problem in either direction matters: $H_A : \mu \neq 12$, $H_0 : \mu = 12$, with $\mu_0 = 12$ oz: **two-sided**.

# Measuring "how surprising": the test statistic

For the taste test we could list every possible score and add up the binomial probabilities. With hundreds or thousands of observations that gets impractical, but we already have a better tool: the **sampling distribution** and the **central limit theorem** from Week 8.

Recall what the CLT told us about $\hat{p}$: over all possible samples of size $n$, the values of $\hat{p}$ pile up in a roughly normal shape, centered at the true $p$, with standard error $SE = \sqrt{p(1-p)/n}$. In a significance test we **assume $H_0$ is true**, so the true $p$ is $p_0$. That pins down the whole pile:

$$\text{if } H_0 \text{ is true:} \qquad \hat{p} \;\approx\; N\left(p_0,\; \sqrt{\frac{p_0(1-p_0)}{n}}\right)$$

Notice where every number comes from. The **center** $p_0$ comes from $H_0$, not from the data. The standard error also uses $p_0$, because we are describing the world where $H_0$ is true. The data enter only once, as the single $\hat{p}$ we observed, which we then place on that curve.

To see how far out on the curve it lands, we turn it into a $z$-score, exactly as in Weeks 6 and 9 (observation minus mean, divided by standard deviation). This $z$-score is the **test statistic**:

$$z_{obs} = \frac{\hat{\theta} - \theta_0}{SE} \qquad \text{for a proportion:} \qquad z_{obs} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

It measures **how many standard errors the estimate is from the value $H_0$ claims**. If $H_0$ is true, $z_{obs}$ comes from the standard normal $N(0,1)$, so values near 0 are ordinary and values beyond about $\pm 2$ are rare. The P-value is then an area under the standard normal curve, found with the $z$-table just as in Week 9.

This is the same machinery as a confidence interval, pointed in a different direction:

| | Confidence interval | Significance test |
|---|---|---|
| Question | What is the parameter, give or take? | Is the claim $\theta = \theta_0$ believable? |
| Normal curve centered at | the estimate $\hat{\theta}$ | the claimed value $\theta_0$ |
| Uses | $\hat{\theta} \pm z \times SE$ | $z_{obs} = (\hat{\theta} - \theta_0)/SE$ |
| Result | a range of believable values for $\theta$ | a P-value, then reject or do not reject $H_0$ |

## Example: has the candidate's support dropped?

It is midterm season (Election Day is Tuesday, November 3). Suppose the polling averages put Candidate X in a Senate race at **52%** support among likely voters. Your campaign suspects X's support is actually lower. You run your own poll of a random sample of $n = 1{,}000$ likely voters, and **490** say they will vote for X: $\hat{p} = 0.490$.

49% is lower than 52%. But Week 8 taught us that two polls of the same state give different answers just because they reach different people. Is a 3-point gap **bigger than ordinary poll-to-poll variation**? That is exactly a significance test.

1. **Assumptions.** A random sample of likely voters, and a large enough sample: $np_0 = 1000(0.52) = 520$ and $n(1-p_0) = 480$, both well above 15.
2. **Hypotheses.** $p$ = the proportion of **all** likely voters who support X. We suspect it is lower than the polls say, so $H_A : p < 0.52$ and $H_0 : p = 0.52$, with $p_0 = 0.52$: **lower-tailed**.
3. **Test statistic.** If $H_0$ is true, polls of 1,000 voters give values of $\hat{p}$ centered at 0.52 with

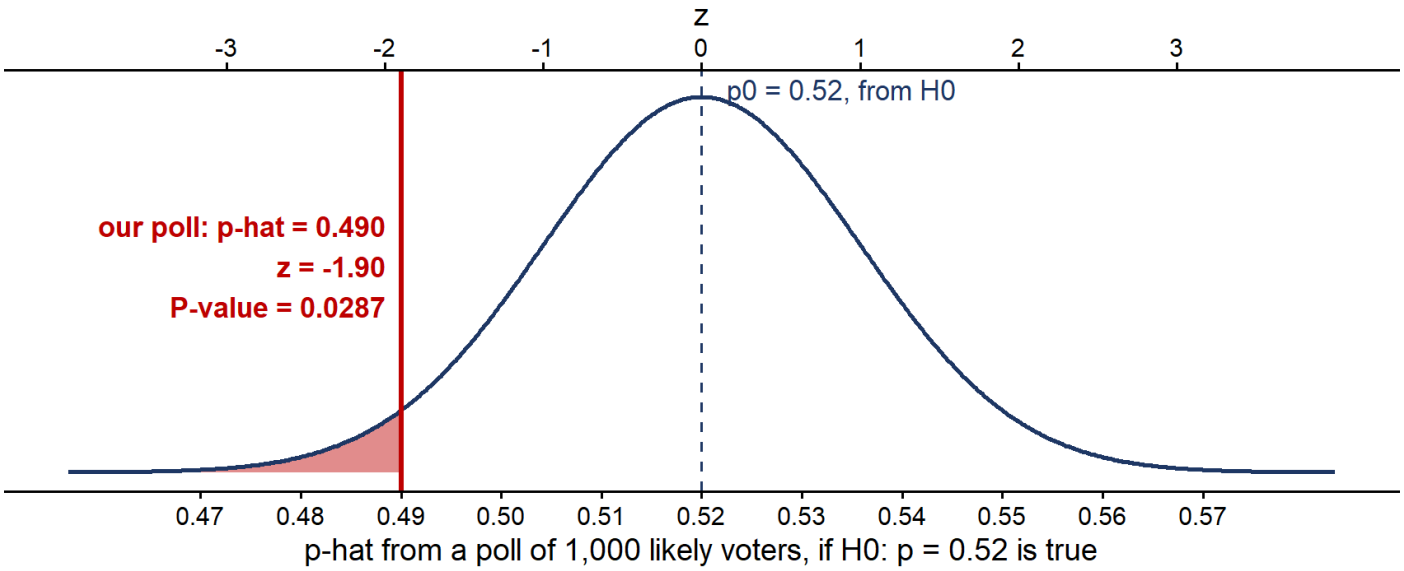
$$SE = \sqrt{\frac{0.52 \times 0.48}{1000}} = \sqrt{0.0002496} \approx 0.0158$$

   so

$$z_{obs} = \frac{0.490 - 0.52}{0.0158} \approx -1.90$$

   Our poll is 1.9 standard errors below what $H_0$ predicts.
4. **P-value.** $P(Z \le -1.90 \mid H_0 \text{ true}) = 0.0287$, from the $z$-table. If X's support really were 52%, only about 3 polls in 100 would come out this low.
5. **Decision.** $0.0287 < 0.05$: **reject** $H_0$. The poll is evidence that X's support is below 52%.

Here is the same reasoning as a picture. The curve shows where $\hat{p}$ from a poll of 1,000 voters would land **if $H_0$ were true**; the red line is where ours landed, and the shaded area is the P-value.

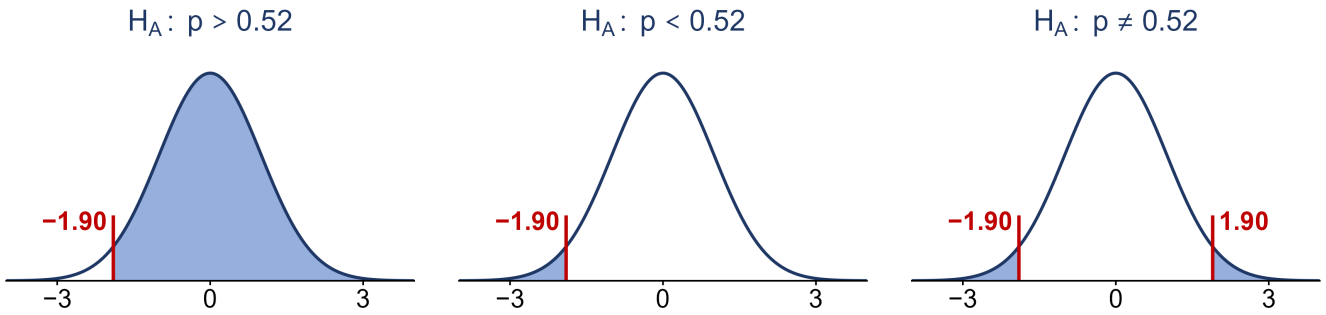



Read the two axes together: the top axis is in the units of the problem (proportions of voters), the bottom axis is the same picture in standard errors. The test statistic is just the poll's position on the bottom axis.

**A different claim, the same poll.** The headline question on election night is whether X is **losing**, i.e. below 50%. That is a different $H_0$: $H_0 : p = 0.5$, $H_A : p < 0.5$. Now $SE = \sqrt{0.25/1000} \approx 0.0158$ and $z_{obs} = (0.490 - 0.5)/0.0158 \approx -0.63$, so the P-value is $P(Z \leq -0.63) = 0.2643$. **Do not reject $H_0$.** The same 49% that is good evidence against "52%" is weak evidence against "50%", because 0.490 sits only 0.63 standard errors below 0.5. A test always answers a question about one specific value $\theta_0$: change $\theta_0$ and you change the question. This is why poll reports say a race is "within the margin of error".

## Which tail is the P-value?

"At least as extreme" always means "at least as far in the direction $H_A$ points to". So the shape of $H_A$ decides which area of the curve is the P-value. To see how much it matters, take our poll ($z_{obs} = -1.90$ against $p_0 = 0.52$) and work out the P-value for all three alternatives:




- **Lower-tailed**, $H_A : p < 0.52$: the area **below** $z_{obs}$, $P(Z \leq -1.90 \mid H_0 \text{ true}) = 0.0287$, read straight from the table.
- **Two-sided**, $H_A : p \neq 0.52$: a poll far out in **either** direction counts as extreme, so the P-value is **both** tails, twice the one-tail area: $2 \times 0.0287 = 0.0574$.
- **Upper-tailed**, $H_A : p > 0.52$: the area **above** $z_{obs}$, $P(Z \geq -1.90 \mid H_0 \text{ true}) = 1 - 0.0287 = 0.9713$. A poll **below** 52% is no evidence at all that support is **above** 52%, and the huge P-value says so.

At $\alpha = 0.05$, only the lower-tailed test rejects $H_0$. The two-sided test, with the same data, does not (0.0574): it spreads its doubt over both directions, so it needs a little more evidence. This is why $H_A$ has to come from the question before the poll is run. Choosing the tail after seeing which way the poll went would make any result look more convincing than it is.

# Example: do coins land the way they started?

A physics model of coin tossing (Diaconis, Holmes and Montgomery, 2007) predicts that a flipped coin lands on the **same side it started on** a little more than half the time, about 51%, because a real flip wobbles. Bartoš et al. (2025) tested it: 48 people flipped coins 350,757 times, recording the side facing up before and after every flip (and filming it).

| Who flipped | Flips |
|---|---|
| students, for a thesis project | 75,036 |
| coin-flipping marathons (up to 12 hours) | 203,440 |
| volunteers recruited online | 72,281 |
| **total** | **350,757** |

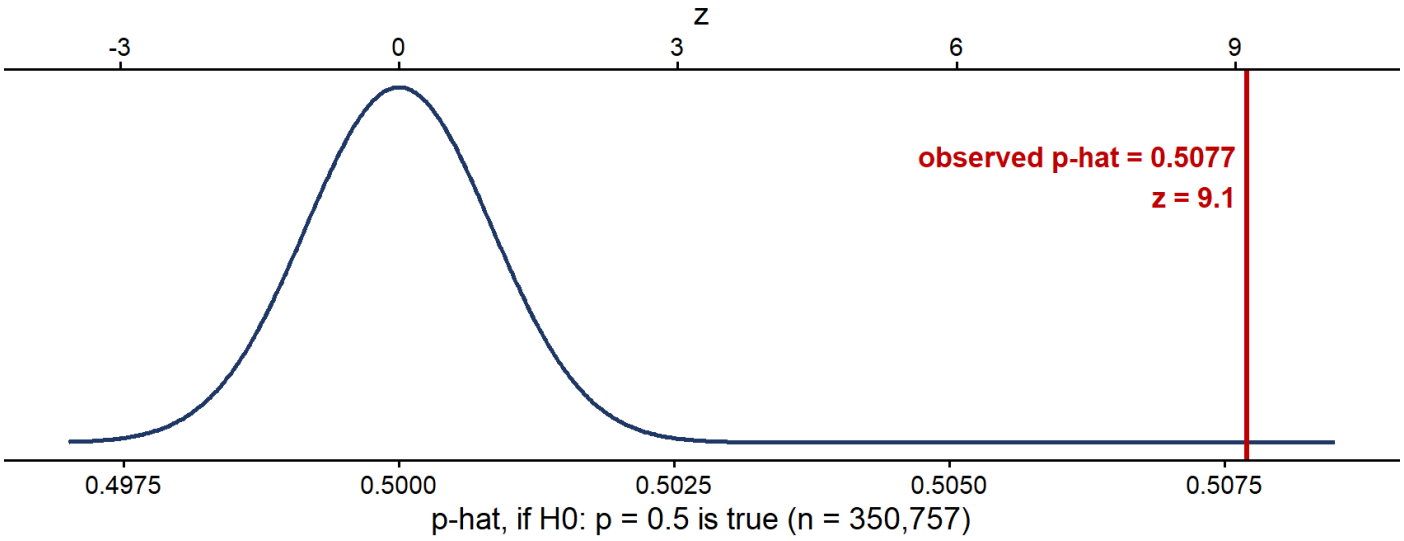
$X$ = 1 if a flip lands on the side it started on, and 0 otherwise. In all, **178,079** of the $n = 350{,}757$ flips did. Test the model's prediction at $\alpha = 0.05$, using the five steps.

1. **Assumptions.** The flips are a random sample of how people flip coins, and $n$ is large: $np_0$ and $n(1-p_0)$ are both about 175,000, far more than enough for the normal curve to apply.
2. **Hypotheses.** $p$ = the chance a flip lands on its starting side. The model predicted the direction before any flips were made, so $H_A : p > 0.5$ and $H_0 : p = 0.5$, with $p_0 = 0.5$: **upper-tailed**.
3. **Test statistic.** $\hat{p} = 178{,}079/350{,}757 \approx 0.5077$ and $SE = \sqrt{0.5 \times 0.5/350{,}757} \approx 0.000844$, so

$$z_{obs} = \frac{0.5077 - 0.5}{0.000844} \approx 9.1$$

   The estimate sits about **9 standard errors** above $p_0$.
4. **P-value.** $P(Z \geq 9.1 \mid H_0 \text{ true}) \approx 0$. The $z$-table stops at 3.49, and 9.1 is far beyond it.
5. **Decision.** The P-value is far below 0.05: **reject** $H_0$. The data are strong evidence that flips land on their starting side more than half the time, just as the model predicted.

Here is what step 3 looks like on the sampling distribution. The curve is where $\hat{p}$ would land if $H_0$ were true; the red line is where it actually landed.

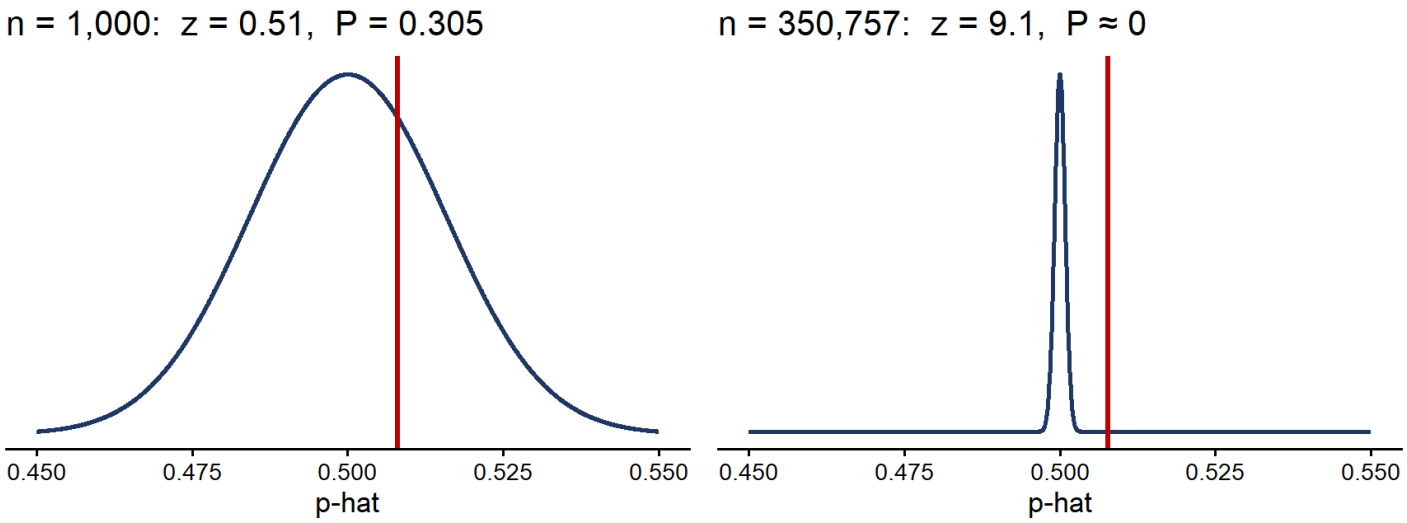



# Significant is not the same as important

A 50.8% same-side rate instead of 50% is about **1 extra same-side landing in every 130 flips**. Nobody could notice that in a coin toss before a football game. So how can it be "significant"?

Because significance depends on the **sample size**. The standard error has $\sqrt{n}$ on the bottom, so with 350,757 flips the curve of believable $\hat{p}$ values is extremely narrow, and even a tiny difference lands far out in the tail. Suppose instead only 1,000 flips had been made, with the same 50.8% (508

same-side). Then $SE = \sqrt{0.25/1000} \approx 0.0158$, $z_{obs} = (0.508 - 0.5)/0.0158 \approx 0.51$, and the P-value is $1 - 0.6950 = 0.3050$: **not** significant.




Same estimate, opposite conclusions. "Statistically significant" means **"probably not chance"**. It does not mean "big" or "important". Whether a difference matters is a question about the context, and a confidence interval (which shows **how big** the effect is) answers it better than a P-value does.

# The five steps of a significance test

Every test in the rest of the course follows the same five steps:

1. **Assumptions.** The data come from a random sample or a randomized experiment, plus any conditions on sample size or on the shape of the population.
2. **Hypotheses.** $H_0 : \theta = \theta_0$ and $H_A$ (upper-tailed, lower-tailed or two-sided), written before looking at the data.
3. **Test statistic.** $z_{obs} = (\hat{\theta} - \theta_0)/SE$, computed as if $H_0$ were true: how many standard errors the estimate is from $\theta_0$.
4. **P-value.** The probability, if $H_0$ is true, of a test statistic at least as extreme as $z_{obs}$, in the tail or tails that $H_A$ points to.
5. **Decision.** If the P-value $< \alpha$, reject $H_0$ (statistically significant); otherwise do not reject $H_0$. Then say what it means in the context of the question.

"Do not reject $H_0$" is not the same as "accept $H_0$". Not rejecting $H_0$ does not prove it true; the data just are not strong enough evidence against it.

## Practice: P-values and decisions

For each test, write the P-value as a probability given $H_0$ is true. Then use its value to make a decision.

| Test | Hypotheses | Test statistic | $\alpha$ | P-value |
|---|---|---|---|---|
| (a) | $H_0 : p = 0.40$, $H_A : p > 0.40$ | $z_{obs} = 1.52$ | 0.05 | 0.0643 |
| (b) | $H_0 : \mu = 12$, $H_A : \mu < 12$ | $t_{obs} = -2.30$, $df = 15$ | 0.01 | 0.0181 |
| (c) | $H_0 : \mu = 50$, $H_A : \mu \neq 50$ | $z_{obs} = -2.10$ | 0.05 | 0.0358 |

*Answers:*

- a. $P(Z \geq 1.52 \mid H_0 \text{ true}) = 0.0643 \geq 0.05$: do not reject $H_0$.
- b. $P(T \leq -2.30 \mid H_0 \text{ true}) = 0.0181 \geq 0.01$: do not reject $H_0$. (It would be significant at $\alpha = 0.05$.)
- c. $P(Z \leq -2.10 \text{ or } Z \geq 2.10 \mid H_0 \text{ true}) = 2(0.0179) = 0.0358 < 0.05$: reject $H_0$.

The same five steps test claims about a population proportion $p$ and about a population mean $\mu$.

# Testing a population proportion

When the variable is categorical and the claim is about the proportion of successes, $H_0 : p = p_0$, the five steps become:

1. **Assumptions.** A random sample, and a large enough sample: $np_0 \geq 15$ and $n(1 - p_0) \geq 15$.
2. **Hypotheses.** $H_0 : p = p_0$ against $H_A : p > p_0$, $p < p_0$ or $p \neq p_0$.
3. **Test statistic.**

$$z_{obs} = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

   The standard error uses $p_0$, not $\hat{p}$, because the whole calculation assumes $H_0$ is true. (A confidence interval uses $\hat{p}$, because there is no $p_0$ to assume.)
4. **P-value.** Upper-tailed: $P(Z \geq z_{obs} \mid H_0 \text{ true})$. Lower-tailed: $P(Z \leq z_{obs} \mid H_0 \text{ true})$. Two-sided: double the tail beyond $z_{obs}$.
5. **Decision.** Compare the P-value with $\alpha$, then say what it means in context.

The coin-flip study above is one of these tests. Here is a second one, from the textbook.

## Example: cell phones and brain cancer

*From OpenStax Introductory Statistics 2e, Example 9.21.*

Almost everyone carries a cell phone, and most of us hold it against our heads every day. That has led to a worry that comes up again and again: can the radio waves from a phone cause brain cancer?

Brain cancer is rare. Among people who do **not** use cell phones, about 0.034% develop it, which is 34 people in every 100,000. If phones raise the risk, then the rate among cell phone users should be higher than 0.034%.

To check, researchers follow a very large group of **420,019 cell phone users**. Over the course of the study, **172** of them develop brain cancer.

If phones made no difference, we would expect about $420{,}019 \times 0.00034 \approx 143$ of these users to develop brain cancer. We saw 172, which is more. But is 172 far enough above 143 to rule out chance? Or is it the kind of number a group this size could produce even if phones are harmless?

The stakes are high. Announcing that cell phones cause cancer when they do not would frighten a great many people and send research and regulation chasing a false cause. So before making that claim we want very strong evidence, and we use a strict significance level of $\alpha = 0.005$ instead of the usual 0.05.

**Question:** At $\alpha = 0.005$, do the data show that cell phone users develop brain cancer at a higher rate than non-users?

1. **Assumptions.** We treat the 420,019 users as a random sample of cell phone users. Each user either develops brain cancer or does not, and the sample is large enough: $np_0 = 420{,}019(0.00034) = 142.8 \geq 15$ and $n(1-p_0) = 420{,}019(0.99966) = 419,876 \geq 15$. The first number is the "about 143" expected cases from the story.
2. **Hypotheses.** The parameter is $p$, the proportion of **all** cell phone users who develop brain cancer. The null value is the non-user rate, $p_0 = 0.034\% = 0.00034$. $H_0 : p = 0.00034$ (phones make no difference) and $H_A : p > 0.00034$ (users get brain cancer at a higher rate). The claim is about a **higher** rate, so the test is **upper-tailed**.
3. **Test statistic.** The sample proportion is $\hat{p} = 172/420{,}019 \approx 0.0004095$, or about 0.041%. The standard error uses $p_0$, because we work as if $H_0$ is true:

$$SE = \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{0.00034 \times 0.99966}{420{,}019}} \approx 0.00002845$$

$$z_{obs} = \frac{\hat{p} - p_0}{SE} = \frac{0.0004095 - 0.00034}{0.00002845} \approx 2.44$$

The users' cancer rate is 2.44 standard errors above the non-user rate.

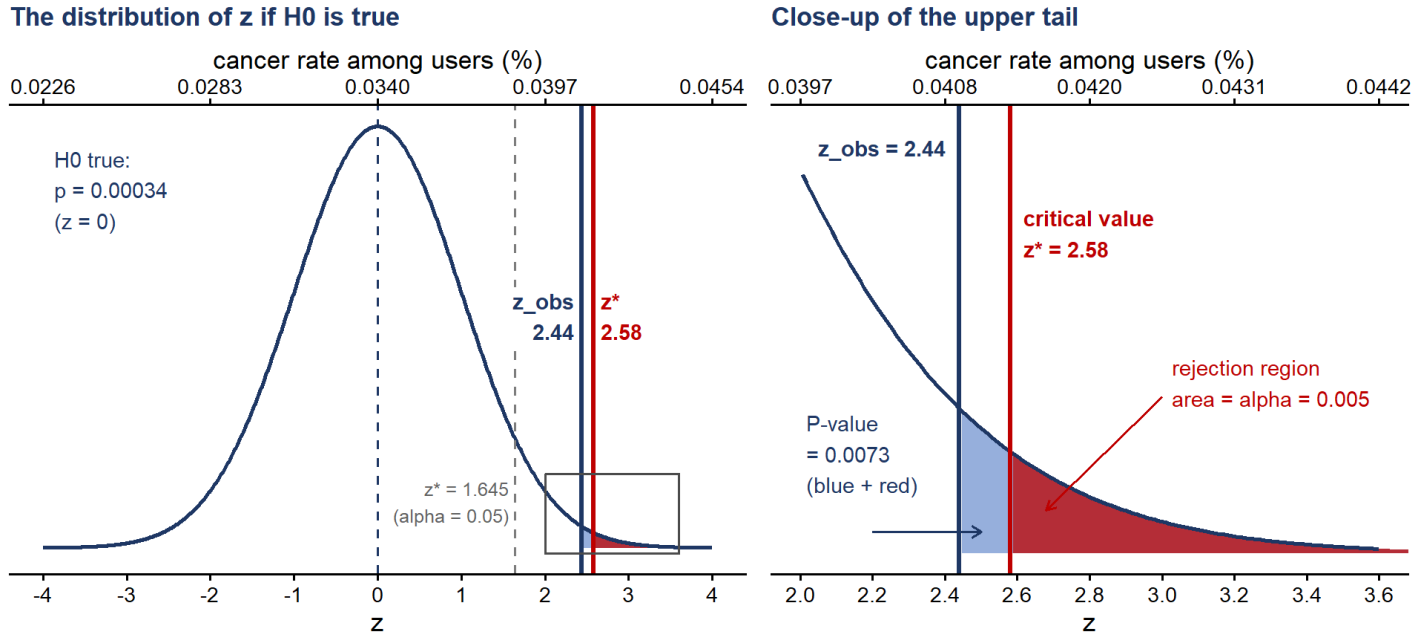
4. **P-value.** Upper-tailed, so we want the area to the right of $z_{obs}$:

$$P(Z \geq 2.44 \mid H_0 \text{ true}) = 1 - 0.9927 = 0.0073$$

In words: if phones made no difference, a study this size would see a cancer rate this far above 0.034% only about 73 times in 10,000.

5. **Decision.** $0.0073 \geq 0.005$: **do not reject** $H_0$. At the 0.5% level, the data are not strong enough evidence that cell phone users develop brain cancer at a higher rate than non-users. This does **not** prove phones are safe. The evidence just does not clear the high bar we set.

Here is the whole test as a picture. The curve is the distribution of $z$ **if** $H_0$ **is true** (phones make no difference). The **rejection region** is the upper 0.5% of that curve, everything at or beyond the **critical value** $z^* = 2.58$, so $P(Z \geq 2.58 \mid H_0 \text{ true}) = 0.005 = \alpha$. Our test statistic, $z_{obs} = 2.44$, lands just short of it. The area beyond $z_{obs}$ is the P-value, 0.0073, and it is a little bigger than $\alpha$. That is the decision rule in picture form: $z_{obs}$ inside the rejection region exactly when the P-value is less than $\alpha$.




The grey dashed line shows the same test at the usual $\alpha = 0.05$: the critical value would move left to 1.645, and $z_{obs} = 2.44$ would fall inside the rejection region.

Why such a small $\alpha$? Rejecting $H_0$ when it is true would wrongly blame cell phones for cancer, a claim with serious consequences, so we ask for very strong evidence before making it. Notice that the P-value of 0.0073 would have been significant at the usual $\alpha = 0.05$. The decision depends on the cutoff, which is why $\alpha$ is chosen **before** the data are collected.

## Assumptions for a population proportion

Intervals and tests for $p$ both need a random sample, with each individual counted as a success or a failure. They differ in the sample-size check:

- **Confidence interval for $p$:** $n\hat{p} > 5$ and $n(1-\hat{p}) > 5$, checked with $\hat{p}$, because $p$ is unknown.
- **Significance test for $p$:** $np_0 \geq 15$ and $n(1-p_0) \geq 15$, checked with $p_0$, because the test assumes $H_0$ is true.

Either way, the check is there to make the sampling distribution of $\hat{p}$ close to normal, so that the areas from the $z$-table are right.

## Try it out

- A friend claims they can predict coin flips. In a test of 10 flips, they call $\mathbf{7}$ correctly. Write the hypotheses and find the P-value. Is the result significant at $\alpha = 0.05$? *Answer:* $H_A : p > 0.5$, $H_0 : p = 0.5$ (upper-tailed). Under $H_0$, $X \sim \text{Binom}(10, 0.5)$, so P-value $= P(X \geq 7 \mid H_0 \text{ true}) = (120 + 45 + 10 + 1)/1024 = 176/1024 \approx 0.172$. $0.172 \geq 0.05$: do not reject $H_0$. Seven of ten is quite common for a guesser.
- Write $H_0$ and $H_A$, give $\theta_0$, and name the type of test.
  - A coffee shop says its large cup holds 16 oz on average. Customers suspect it holds less. *Answer:* $H_0 : \mu = 16$, $H_A : \mu < 16$, $\mu_0 = 16$ oz, lower-tailed.
  - A website's sign-up rate used to be 12%. After a redesign, has it changed? *Answer:* $H_0 : p = 0.12$, $H_A : p \neq 0.12$, $p_0 = 0.12$, two-sided.
- A test statistic comes out to $z_{obs} = -2.10$. Find the P-value if $H_A$ is lower-tailed, and if it is two-sided. What is the decision for each at $\alpha = 0.05$? At $\alpha = 0.01$? *Answer:* lower-tailed: $P(Z \leq -2.10) = 0.0179$; two-sided: $2 \times 0.0179 = 0.0358$. Both are below 0.05, so reject $H_0$ either way at $\alpha = 0.05$. Neither is below 0.01, so do not reject $H_0$ at $\alpha = 0.01$.
- *(The Lecture 23 practice problem.)* Last year 30% of students skipped breakfast. This fall, campus dining asks a random sample of 400 students, and 136 say they skip it. Has the proportion gone up? Use $\alpha = 0.05$. *Answer:* $np_0 = 120$ and $n(1-p_0) = 280$, both at least 15. $H_0 : p$

$= 0.30$, $H_A : p > 0.30$ (upper-tailed). $\hat{p} = 136/400 = 0.34$, $SE = \sqrt{0.30 \times 0.70/400}$ $\approx 0.0229$, $z_{obs} = 0.04/0.0229 \approx 1.75$. P-value $= 1 - 0.9599 = 0.0401 < 0.05$: reject $H_0$. The proportion of students skipping breakfast has gone up.

# Testing a population mean

When the variable is quantitative and the claim is about its mean, $H_0 : \mu = \mu_0$, the five steps become:

1. **Assumptions.** A random sample, from a population that is normal or close to it. With $n \geq 30$ the Central Limit Theorem makes $\bar{x}$ close to normal anyway.
2. **Hypotheses.** $H_0 : \mu = \mu_0$ against $H_A : \mu > \mu_0$, $\mu < \mu_0$ or $\mu \neq \mu_0$.
3. **Test statistic.** Just as for confidence intervals, which one depends on whether $\sigma$ is known:

$$\sigma \text{ known: } z_{obs} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \qquad \sigma \text{ unknown: } t_{obs} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1$$

4. **P-value.** The same tails as for $p$. Upper-tailed: $P(T \geq t_{obs} \mid H_0 \text{ true})$; lower-tailed: $P(T \leq t_{obs} \mid H_0 \text{ true})$; two-sided: double the tail beyond $t_{obs}$. With $z_{obs}$, read it from the $z$-table as before.
5. **Decision.** Compare the P-value with $\alpha$, then say what it means in context.

**P-values from the $t$-table.** The course $t$-table (misc/t_table.html) lists critical values, not areas, so it gives the P-value only between two bounds. In row $df$, find the two values that $|t_{obs}|$ falls between. The P-value is between the upper-tail areas at the top of those two columns (for a two-sided test, use the two-tail areas). Software gives the exact value.

## Example: the first statistics test

*OpenStax Introductory Statistics 2e, Example 9.16.* Statistics students believe the mean score on the first statistics test is 65. An instructor thinks it is higher. He samples ten students' scores, 65, 65, 70, 67, 66, 63, 63, 68, 72, 71, which come from a normal distribution. Use $\alpha = 0.05$.

1. **Assumptions.** $n = 10$ is small, but the scores are normal, so the test works.
2. **Hypotheses.** $\mu$ = the mean score on the first test. $H_0 : \mu = 65$ and $H_A : \mu > 65$: **upper-tailed**.
3. **Test statistic.** $\bar{x} = 67$ and $s = 3.197$. $\sigma$ is unknown, so use $t$: $SE = 3.197/\sqrt{10} \approx 1.011$ and

$$t_{obs} = \frac{67 - 65}{1.011} \approx 1.98, \qquad df = 10 - 1 = 9$$

4. **P-value.** In row 9 of the $t$-table, $1.833 < 1.98 < 2.262$. Those columns have upper-tail areas 0.05 and 0.025, so $0.025 < \text{P-value} < 0.05$. (Software gives 0.0395.)
5. **Decision.** The P-value is below $\alpha = 0.05$: **reject** $H_0$. The data are convincing evidence that the mean score on the first test is higher than 65.

## Practice: a proportion and a mean

- *(Lecture 24 practice; OpenStax Example 9.17.)* Joon believes 50% of first-time brides in the United States are younger than their grooms. She samples 100 first-time brides, and 53 say they are younger than their grooms. Is the percentage different from 50%? Use $\alpha = 0.01$. *Answer:* $np_0 = n(1-p_0) = 50$, both at least 15. $H_0 : p = 0.50$, $H_A : p \neq 0.50$ (two-sided). $\hat{p} = 0.53$, $SE = \sqrt{0.50 \times 0.50/100} = 0.05$, $z_{obs} = 0.03/0.05 = 0.60$. P-value $= 2 \times P(Z \geq 0.60 \mid H_0 \text{ true}) = 2(1 - 0.7257) = 0.5486 \geq 0.01$: do not reject $H_0$. The data are not convincing evidence that the percentage differs from 50%.
- *(Lecture 24 practice; OpenStax Example 9.14.)* Jeffrey's mean time for the 25-yard freestyle is 16.43 seconds, with $\sigma = 0.8$ seconds. In new goggles he swims it 15 times, with a mean of 16 seconds. Do the goggles make him faster? Swim times are normal; use $\alpha = 0.05$. *Answer:* $H_0 : \mu = 16.43$, $H_A : \mu < 16.43$ (lower-tailed: faster means less time). $\sigma$ is known, so use $z$. $SE = 0.8/\sqrt{15} \approx 0.2066$, $z_{obs} = (16 - 16.43)/0.2066 \approx -2.08$. P-value $= P(Z \leq -2.08 \mid H_0 \text{ true}) = 0.0188 < 0.05$: reject $H_0$. Jeffrey swims faster with the goggles.

# Sources

- Coin flips: Bartoš, F., et al. (2025), "Fair coins tend to land on the same side they started: Evidence from 350,757 flips", *Journal of the American Statistical Association* 120(552), 2118–2127 (preprint on arXiv (https://arxiv.org/abs/2310.04153)). The physics model: Diaconis, P., Holmes, S., and Montgomery, R. (2007), "Dynamical bias in the coin toss", *SIAM Review* 49(2), 211–235.
- The taste test, Candidate X's poll, the chip bag, the coffee shop and the website are made-up examples.
- Cell phones and brain cancer: OpenStax *Introductory Statistics 2e*, Example 9.21. The first statistics test, the first-time brides and the goggles are Examples 9.16, 9.17 and 9.14. The breakfast survey is made up.
- Textbook: OpenStax *Introductory Statistics 2e*, Chapter 9, Sections 9.1–9.6.